

Ameritrade Case

Questions for Write-Up and Class Discussion

Read the case and answer the following questions.

Note: A spreadsheet posted on Canvas contains all of the data in Exhibits 1 – 6 of the case.

1. The title suggests that the cost of capital is important here. Concisely describe what cost of capital means. It is the cost of what to whom? Why is it important? What factors determine the cost of capital?
2. Ameritrade has a short history of trading, so its equity beta cannot be computed precisely using its own historical data. Exhibit 4 provides some choices for comparable firms. Which of these firms do you think are appropriate to use as comparables to determine the beta of Ameritrade's planned advertising and technology investments? (i.e., which firms qualify in Step 1 of our five steps)
3. Using the stock price and return data in Exhibits 5 and 6, estimate the CAPM equity beta for firms (i.e., keep firms with usable equity beta in Step 2 of our five steps)

➤ Hint: To calculate returns when there is an x to y stock split, use

$$r_t = \frac{\left(\frac{x}{y}\right)(p_t + d_t) - p_{t-1}}{p_{t-1}}$$

- You can check that your returns are right by matching the mean and standard deviation of Charles Schwab's return over the period Jan 1992-Dec 1996 of mean = 3.345% and st. dev. = 11.513%. Also remember that you need to unlever an equity beta to calculate the asset beta.)
4. Estimate the cost of capital for Ameritrade's proposed project. (i.e., apply the remaining steps in the five-step method to arrive at cost of capital)